4.4 Solving Congruences

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Definition: A linear congruence is an equation of the form $ax \equiv b \pmod{m}$ where $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, and x is a variable.

When finding inverses, we try solving a specific linear congruence where $b = 1$.

Theorem: If $a, m \in \mathbb{Z}$, $m > 1$, and $\gcd(a, m) = 1$ then a multiplicative inverse of "a" mod m exists.

If we can solve one linear congruence, it's reasonable to ask if we can solve a system of them!

In the first century, a Chinese mathematician Sun-Tsu asked:

"There are certain things whose number is unknown. When divided by 3, 2 remain; when divided by 5, 3 remain; when divided by 7, 2 remain. What will be the number of things?"

The answer to this question is the solution of the following system.

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \end{cases}$$

The Chinese Remainder Theorem: Let $m_1, \ldots, m_n \in \mathbb{Z}^+$ be pairwise relatively prime (i.e. $\gcd(m_i, m_j) = 1$ for all $i \neq j$).

Let $a_1, \ldots, a_n \in \mathbb{Z}$. Then the system

$\{\ x \equiv a_1 \pmod{m_1}$ has a unique

Let $a_1, \dots, a_n \in \mathbb{Z}$. Then the system

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \quad \vdots \\ x \equiv a_n \pmod{m_n} \end{cases}$$

has a unique solution modulo $m = m_1 \cdot m_2 \cdots m_n$.
(i.e. $0 \le x < m$)

$$x = (a_1 \cdot M_1 \cdot y_1 + \cdots + a_n \cdot M_n \cdot y_n) \bmod m$$

Where $M_i = \frac{m}{m_i}$ and $M_i \cdot y_i \equiv 1 \pmod{m_i}$

i.e. $y_i = M_i^{-1} \pmod{m_i}$.

Proof: (To prove this we must show there is a solution. Then, any other solution is congruent to ours modulo m)

Let $m_1, \dots, m_n \in \mathbb{Z}^+$ be pairwise relatively prime and $a_1, \dots, a_n \in \mathbb{Z}$. Suppose $x \in \mathbb{Z}$ satisfying the system:

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ \quad \vdots \\ x \equiv a_n \pmod{m_n} \end{cases}$$

Let $m = m_1 \cdots m_n$ and define

$$M_i = \frac{m}{m_i} \quad (= m_1 \cdot m_2 \cdots m_{i-1} \cdot m_{i+1} \cdots m_n)$$

for $i \in \{1, 2, \dots, n\}$. Then since $m_1, \dots, m_n$ are relatively prime and $m_i \nmid M_i$, $\gcd(m_i, M_i) = 1$. Then by theorem, there exists a multiplicative inverse of M_i modulo m_i, call it y_i. Then

$$M_i \cdot y_i \equiv 1 \pmod{m_i}.$$

Define $x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_n M_n y_n$.
Consider this equation modulo m_i for each $i \in \{1, 2, \dots, n\}$.

$$x \equiv (a_1 M_1 y_1 + \cdots + a_n M_n y_n) \pmod{m_i}$$

All terms except M_i are multiples of m_i

All terms except M_i are multiples of m_i thus give zero. Then $x \equiv a_i M_i y_i \pmod{m_i}$ but $M_i y_i \equiv 1 \pmod{m_i}$ so we have

$x \equiv a_i \pmod{m_i}$ for every $i \in \{1,2,\dots,n\}$. Therefore this choice of x solves the system.

Showing uniqueness is a good exercise. Problems 29 and 30 guide you through it.

So how do we solve these systems??

Using $x = a_1 M_1 y_1 + \cdots + a_n M_n y_n$ as defined in the proof.

Solve the system.

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 3 \pmod 5 \\ x \equiv 2 \pmod 7 \end{cases}$$

3, 5, 7 <u>are</u> rel. prime!

$m_1 = 3 \quad a_1 = 2$

$m_2 = 5 \quad a_2 = 3$

$m_3 = 7 \quad a_3 = 2$

$m = 3\cdot 5\cdot 7$, $M_1 = 5\cdot 7$, $M_2 = 3\cdot 7$, $M_3 = 3\cdot 5$

$y_1 \cdot M_1 \equiv 1 \pmod{m_1} \Rightarrow y_1 \cdot (5\cdot 7) \equiv 1 \pmod 3$

$(y_1 \cdot 5 \cdot 7) \bmod 3 = (y_1 \bmod 3 \cdot 5 \bmod 3 \cdot 7 \bmod 3) \bmod 3$

$= (y_1 \cdot 2) \bmod 3 \Rightarrow y_1 \cdot 2 \equiv 1 \pmod 3$

or $\boxed{y_1 = 2}$

$y_2 M_2 \equiv 1 \pmod{m_2} \Rightarrow y_2 (3\cdot 7) \equiv 1 \pmod 5$

$\Rightarrow y_2 \cdot 3 \cdot 2 \equiv 1 \pmod 5 \Rightarrow y_2 \equiv 1 \pmod 5$

or $\boxed{y_2 = 1}$

$y_3 M_3 \equiv 1 \pmod{m_3} \Rightarrow y_3 (3\cdot 5) \equiv 1 \pmod 7$

$\Rightarrow y_3 (15) \equiv 1 \pmod 7 \Rightarrow y_3 \equiv 1 \pmod 7$

or $\boxed{y_3 = 1}$

Then $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \bmod m$

$= (2\cdot 5\cdot 7\cdot 2 + 3\cdot 3\cdot 7\cdot 1 + 2\cdot 3\cdot 5\cdot 1) \bmod (3\cdot 5\cdot 7)$

$= (140 + 63 + 30) \bmod 105$

$= (2\cdot5\cdot7\cdot2 + 3\cdot3\cdot7\cdot1 + 2\cdot3\cdot5\cdot1) \bmod (3\cdot5\cdot7)$

$= (140 + 63 + 30) \bmod 105$

$= (35 + 63 + 30) \bmod 105$

$= (98 + 30) \bmod 105 = 128 \bmod 105 \boxed{= 23}$

Check yourself!

$$\begin{cases} 23 \overset{?}{\equiv} 2 \pmod 3 \\ 23 \overset{?}{\equiv} 3 \pmod 5 \\ 23 \overset{?}{\equiv} 2 \pmod 7 \end{cases}$$

It works!

This is the constructive method for solving systems. It is not always the most convenient. There's another way called the back substitution method!

Solve the system

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 3 \pmod 5 \\ x \equiv 2 \pmod 7 \end{cases}$$

$\longrightarrow$ If $x \equiv 2 \pmod 3$, there is an integer u so that $x = 3\cdot u + 2$.

Then $x \equiv 3 \pmod 5 \Rightarrow 3u + 2 \equiv 3 \pmod 5$

$\Rightarrow 3u \equiv 1 \pmod 5$ so $u \equiv 2 \pmod 5$.

Then there is an integer v so that

$u = v\cdot5 + 2 \Rightarrow x = 3u + 2 = 3(v\cdot5 + 2) + 2$

or $x = 15v + 8$.

Then $x \equiv 2 \pmod 7 \Rightarrow 15v + 8 \equiv 2 \pmod 7$

$\Rightarrow 15v + 1 \equiv 2 \pmod 7 \Rightarrow 15v \equiv 1 \pmod 7$

$\Rightarrow v \equiv 1 \pmod 7$. Then there is an integer w so that $v = 7w + 1$.

Then $x = 15v + 8 = 15(7w + 1) + 8 \bmod 105$

$= 105\cdot w + 15 + 8 \bmod 105 \boxed{= 23}$

Solve the system of congruences.

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 1 \pmod 4 \end{cases}$$

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 1 \pmod 4 \\ x \equiv 3 \pmod 5 \end{cases}$$

$x \equiv 2 \pmod 3 \Rightarrow x = 3u + 2$

$x \equiv 1 \pmod 4 \Rightarrow 3u + 2 \equiv 1 \pmod 4$

$\Rightarrow 3u \equiv -1 \pmod 4$

$\Rightarrow 3u \equiv 3 \pmod 4$

$\Rightarrow u \equiv 1 \pmod 4$

$\Rightarrow u = 4v + 1$

So $x = 3u + 2 = 3(4v+1) + 2 = 12v + 5$

$x \equiv 3 \pmod 5 \Rightarrow 12v + 5 \equiv 3 \pmod 5$

$\Rightarrow 2v \equiv 3 \pmod 5$

$\Rightarrow v \equiv 4 \pmod 5$

$\Rightarrow v = 5w + 4$

So $x = 12v + 5 = 12(5w + 4) + 5$

$= 60w + 48 + 5 = 60w + 53$

$m = 3 \cdot 4 \cdot 5 = 60.$

$\boxed{x \equiv 53 \bmod 60}$

$53 \stackrel{?}{\equiv} 2 \pmod 3$ $\quad 53 = 17 \cdot 3 + 2$

$53 \stackrel{?}{\equiv} 1 \pmod 4$ $\quad 53 = 13 \cdot 4 + 1$

$53 \stackrel{?}{\equiv} 3 \pmod 5$ $\quad 53 = 10 \cdot 5 + 3$

Whenever we find a new description of x, it is important that x is still <u>always an integer</u>.

That is why we solve for u and v completely before substituting. Otherwise if we stop too soon, we may accidentally pick up we may accidentally pick up a conflict in data types.

$$\begin{cases} x \equiv 2 \pmod 4 & (1) \\ x \equiv 1 \pmod 5 & (2) \end{cases}$$

To satisfy (1), $x = 2+4a$ for any integer a.

To satisfy (2) also, then $x \equiv 1 \pmod 5$

so $(2+4a) \equiv 1 \pmod 5$

$\rightarrow 4a \equiv -1 \pmod 5 \Rightarrow 4a \equiv 4 \pmod 5$

If we take $4a = 4 + 5b$ for any integer b, then $a = 1 + \frac{5}{4}b$.

Then a is only an integer when b is a multiple of 4

Continuing to solve $4a \equiv 4 \pmod 5$

then $4 \cdot (4a) \equiv 4 \cdot (4) \pmod 5$

$\rightarrow 16a \equiv 16 \pmod 5 \Rightarrow a \equiv 1 \pmod 5$

(since $16 \bmod 5 = 1$).

So $a = 1 + 5b$ for any integer b.

Using $a = 1 + \frac{5}{4}b$

$x = 2 + 4a = 2 + 4(1 + \frac{5}{4}b)$

$= 2 + 4 + 5b$

$x = 6 + 5b$ but

then x is not always a solution to the system!

$b = 1$ $\quad x = 11$

$11 \equiv 3 \pmod 4$ ☹

$11 \equiv 1 \pmod 5$ ✓

Using $a = 1 + 5b$

$x = 2 + 4(1 + 5b) = 2 + 4 + 20b$

$x = 6 + 20b$ so then

for every integer b

$(6 + 20b) \equiv 2 \pmod 4$ ☺

$(6 + 20b) \equiv 1 \pmod 5$ ☺

so $x \equiv 6 \pmod{20}$

__Fermat's Little Theorem__: If $p>0$ is prime and a is an integer not divisible by p ($p \nmid a$ or equivalently $\gcd(a,p)=1$) then

$$\begin{cases} a^{p-1} \equiv 1 \pmod{p} \\ a^{p} \equiv a \pmod{p} \end{cases}$$

So we can simplify powers in congruences by dividing the power by our modulus as long as the modulus is prime!

Find $7^{222} \pmod{11}$

$11 = p$

$$7^{222} = 7^{20\cdot 11+2} = (7^{11})^{20}\cdot 7^2$$

$$\equiv (7)^{20}\cdot 7^2 \pmod{11}$$

$10 = p-1$

$$= 7^{2\cdot 10}\cdot 7^2 = (7^{10})^2\cdot 7^2$$

$$\equiv (1)^2\cdot 7^2 \pmod{11}$$

$$= 49 \equiv 5 \pmod{11}$$

__Bézout's Theorem__: If $a,b \in \mathbb{Z}^+$, then there exists $s,t \in \mathbb{Z}$ such that

$$\gcd(a,b) = \underbrace{a\cdot s + b\cdot t}_{\text{"linear combination of } a \text{ and } b\text{"}}$$

Write $\gcd(16,26)$ as a linear combination of 16 and 26.

$r_1 = 26 \bmod 16 = 10 \longrightarrow 26 = 1\cdot 16 + 10$

$r_2 = 16 \bmod 10 = 6 \longrightarrow 16 = 1\cdot 10 + 6$

$r_3 = 10 \bmod 6 = 4 \longrightarrow 10 = 1\cdot 6 + 4$

$r_4 = 6 \bmod 4 = 2 \longrightarrow 6 = 1\cdot 4 + \boxed{2}$

$r_5 = 4 \bmod 2 = 0 \longrightarrow 4 = 2\cdot 2 + 0$

$\gcd(16,26) = 2$

$6 = 1\cdot 4 + 2 \longleftrightarrow 2 = 6 - 4$

$10 = 1\cdot 6 + 4 \longleftrightarrow 4 = 10 - 6$

$\downarrow$

$2 = 6 - (10-6)$

$$2 = 6 - 10 + 6 = 2\cdot 6 - 10$$

$$16 = 1\cdot 10 + 6 \longleftrightarrow 6 = 16 - 10$$

$$\downarrow$$

$$2 = 2(16 - 10) - 10 = 2\cdot 16 - 2\cdot 10 - 10$$

$$2 = 2\cdot 16 - 3\cdot 10$$

$$26 = 1\cdot 16 + 10 \longleftrightarrow 10 = 26 - 16$$

$$\downarrow$$

$$2 = 2\cdot 16 - 3(26 - 16) = 2\cdot 16 - 3\cdot 26 + 3\cdot 16$$

$$\boxed{2 = 5\cdot 16 - 3\cdot 26}$$